GATE ME 90 Days Challenge

Day 14/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Theory of Machines	Mechanism Analysis	95 days to GATE

How to use this set: 15 questions (7 MCQ, 8 NAT) on Displacement, velocity and acceleration analysis of plane mechanisms (four-bar, slider-crank, quick-return), Grashof's law and inversions, instantaneous centres and Kennedy's theorem, and the Coriolis component. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

A four-bar chain ABCD has AB = 25 mm, BC = 60 mm, CD = 45 mm and DA = 55 mm. To obtain a double-rocker mechanism, the link to be fixed is

(A) AB
(B) BC
(C) CD
(D) DA

Q2 NAT

A planar mechanism has 8 links (including the frame), 10 binary turning pairs and 1 higher pair (cam contact). Its degree of freedom by the Kutzbach criterion is _____

Q3 MCQ

A planar mechanism has a total of 28 instantaneous centres. The number of links in it (including the frame) is

(A) 7
(B) 8
(C) 14
(D) 56

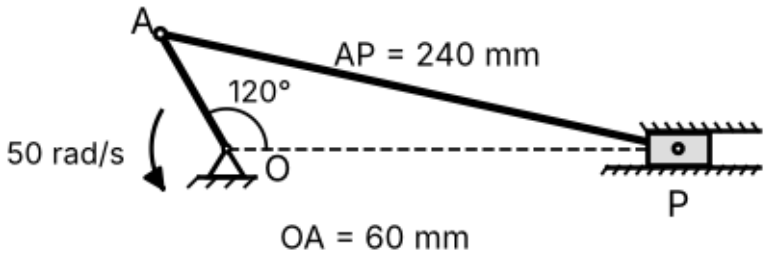
Q4 MCQ

Which of the following is NOT an inversion of the double slider-crank chain?

(A) Elliptical trammel
(B) Scotch yoke
(C) Oldham's coupling
(D) Whitworth quick-return mechanism

Q5 NAT

In the slider-crank mechanism shown, crank OA = 60 mm rotates at 50 rad/s and the connecting rod AP = 240 mm. When the crank is at 120° from the inner dead centre, the velocity of the slider P (in m/s) is _____



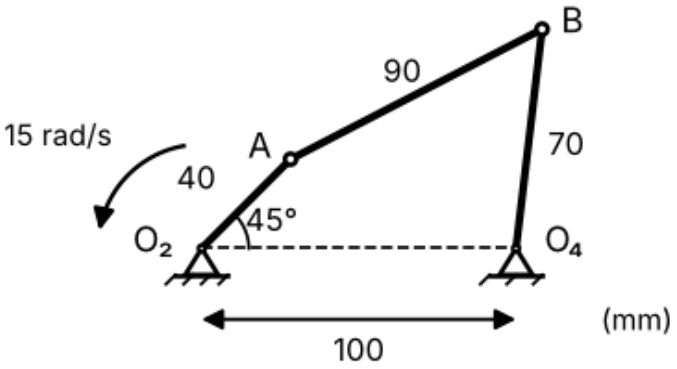
Q6 MCQ

In a slider-crank mechanism the crank is 100 mm, the connecting rod 500 mm and the crank turns uniformly at 40 rad/s. At the inner dead centre, the angular velocity of the connecting rod is

(A) 8 rad/s
(B) 40 rad/s
(C) 200 rad/s
(D) zero

Q7 NAT

In the four-bar mechanism shown (dimensions in mm), the crank O_2A rotates at 15 rad/s counter-clockwise. Using the instantaneous centre method, the angular velocity of the rocker O_4B (in rad/s) at this position is _____



Q8 MCQ

A slider moves radially outward along a link that rotates clockwise at constant speed. The Coriolis component of the slider's acceleration is directed

(A) perpendicular to the link, obtained by rotating the outward sliding velocity 90° clockwise
(B) perpendicular to the link, obtained by rotating the outward sliding velocity 90° counter-clockwise
(C) along the link, outward
(D) along the link, toward the centre of rotation

Q9 NAT

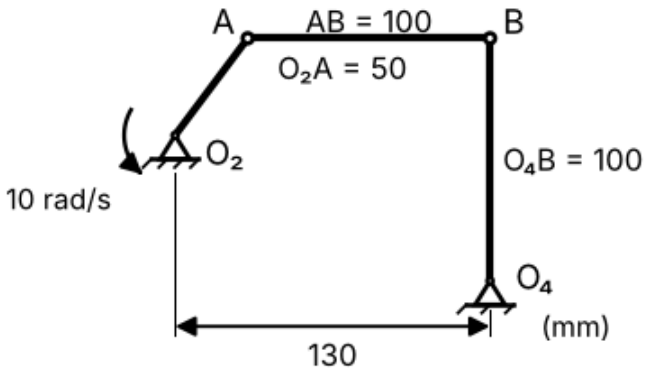
In a crank and slotted-lever quick-return mechanism, the distance between the fixed centres is 240 mm. For the ratio of the cutting-stroke time to the return-stroke time to be 1.5, the length of the crank (in mm) must be _____

Q10 NAT

A horizontal engine has a 75 mm crank and a 375 mm connecting rod, and runs at 900 rpm. The magnitude of the piston's acceleration at the outer dead centre (in m/s²) is _____

Q11 NAT

In the four-bar mechanism shown, O_2 is at the origin, A is at (30, 40) mm, B is at (130, 40) mm and O_4 is at (130, −60) mm. The crank O_2A rotates counter-clockwise at a uniform 10 rad/s. The magnitude of the angular acceleration of the rocker O_4B (in rad/s²) is _____



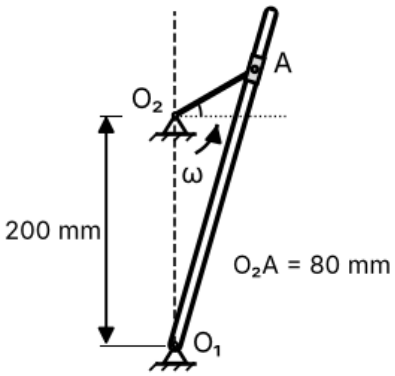
Q12 MCQ

A crank-rocker mechanism has crank 30 mm, coupler 90 mm, rocker 80 mm and fixed link 100 mm. The minimum transmission angle during a revolution of the crank is nearly

(A) 48.2°
(B) 99.6°
(C) 41.8°
(D) 60.0°

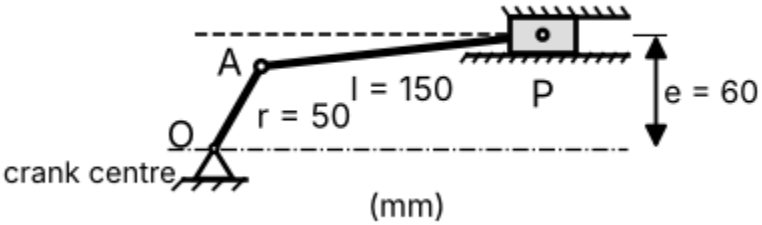
Q13 NAT

In the crank and slotted-lever mechanism shown, the crank O_2A = 80 mm rotates counter-clockwise at 25 rad/s, and the lever pivot O_1 is 200 mm vertically below O_2. When the crank is 30° above the horizontal, the angular velocity of the slotted lever (in rad/s) is _____



Q14 MCQ

In the offset slider-crank mechanism shown, crank r = 50 mm, connecting rod l = 150 mm and the line of stroke is offset by e = 60 mm from the crank centre. The stroke of the slider is nearly



(A) 110.8 mm
(B) 100.0 mm
(C) 90.8 mm
(D) 120.0 mm

Q15 NAT

A slider-crank mechanism has crank OA = 50 mm and connecting rod AP = 200 mm. The crank rotates at 20 rad/s. When the crank is at 45° from the inner dead centre, the speed of the midpoint M of the connecting rod (in m/s) is _____ (round off to 3 decimal places)

Answer Key

Q	Type	Answer
Q1	MCQ	(C) CD
Q2	NAT	0
Q3	MCQ	(B) 8
Q4	MCQ	(D) Whitworth quick-return mechanism
Q5	NAT	2.27 (accept 2.24 to 2.29)
Q6	MCQ	(A) 8 rad/s
Q7	NAT	3.15 (accept 3.12 to 3.18)
Q8	MCQ	(A) perpendicular to the link, obtained by rotating the outward sliding velocity 90° clockwise
Q9	NAT	74.16 (accept 73.8 to 74.5)
Q10	NAT	532.96 (accept 530 to 536)
Q11	NAT	39 (accept 38.6 to 39.4)
Q12	MCQ	(A) 48.2°
Q13	NAT	5.77 (accept 5.71 to 5.83)
Q14	MCQ	(A) 110.8 mm
Q15	NAT	0.848 (accept 0.839 to 0.856)

Detailed Solutions

Q1 **Answer: (C) CD**

Concept: Grashof's law: if $s + l \le p + q$, at least one link can rotate fully. Which link is fixed then decides the type.

$s + l = 25 + 60 = \mathbf{85}$; $p + q = 45 + 55 = \mathbf{100}$. $85 < 100$, so the chain is Grashof.

Fix a link adjacent to the shortest link (BC or DA) → crank-rocker. Fix the shortest link AB → double-crank. Fix the link **opposite** the shortest (CD) → **double-rocker**.

Exam tip: In a Grashof chain the shortest link is the only one that can rotate fully. Find it first, then read off the type from where the fixed link sits relative to it.

Q2 **Answer: 0**

Concept: Kutzbach (planar): $F = 3(n - 1) - 2j - h$, where j = lower pairs and h = higher pairs.

$F = 3(8 - 1) - 2(10) - 1 = 21 - 20 - 1 = \mathbf{0}$

$F = 0$ means the assembly is a structure: no relative motion is possible.

Exam tip: $F = 1$ is a constrained mechanism, $F = 0$ a structure, $F < 0$ a statically indeterminate (superstructure). Count each ternary joint as two binary joints.

Q3 **Answer: (B) 8**

Concept: With n links there are N = n(n − 1)/2 instantaneous centres, one for each pair of links.

n(n − 1)/2 = 28 → n(n − 1) = 56 → **n = 8** (8 × 7 = 56)

Exam tip: 56 is n(n − 1), not n. Primary ICs are found by inspection (pins, sliders, rolling contacts); the rest come from Kennedy's theorem.

Q4 **Answer: (D) Whitworth quick-return mechanism**

Concept: The double slider-crank chain has two turning pairs and two sliding pairs. Its three inversions are the elliptical trammel, the Scotch yoke and Oldham's coupling.

The Whitworth quick-return mechanism comes from the **single** slider-crank chain, by fixing the crank.

Exam tip: Single slider-crank inversions: engine (frame fixed), Whitworth / rotary engine (crank fixed), oscillating cylinder / slotted lever (connecting rod fixed), hand pump (slider fixed).

Q5 **Answer: 2.27 (accept 2.24 to 2.29)**

Concept: Exact slider-crank velocity: $v_P = \omega r \sin(\theta + \varphi)/\cos\varphi$, where $\sin\varphi = (r/l)\sin\theta$.

n = l/r = 4; $\sin\varphi = \sin 120°/4 = 0.2165$ → **$\varphi = 12.50°$**

$\omega r = 50 \times 0.06 =$ **3.0 m/s**

$v_P = 3.0 \times \sin 132.50°/\cos 12.50° = 3.0 \times 0.7373/0.9763 =$ **2.27 m/s**

Exam tip: The approximate form $\omega r(\sin\theta + \sin 2\theta/2n)$ gives 3.0 × (0.866 − 0.108) = 2.27 m/s here too. For 90° < θ < 180° the sin 2θ term is negative, so the slider is slower than ωr sin θ.

Q6 **Answer: (A) 8 rad/s**

Concept: $\omega_{AP} = \omega\cos\theta/\sqrt{(n^2 - \sin^2\theta)}$, where n = l/r.

At θ = 0: $\omega_{AP} = \omega/n = 40/5 =$ **8 rad/s**

Equivalently, $v_A = 40 \times 0.1 = 4$ m/s is perpendicular to the rod, and the slider is momentarily at rest, so $\omega_{AP} = 4/0.5 = 8$ rad/s.

Exam tip: The rod's angular velocity is greatest at the dead centres (ω/n) and zero at θ = 90°. Don't mix this up with its angular acceleration, which is zero at the dead centres.

Q7 **Answer: 3.15 (accept 3.12 to 3.18)**

Concept: The IC of the coupler (I_{13}) lies where the line O_2A extended meets the line O_4B extended. Every point of AB moves as if rotating about it.

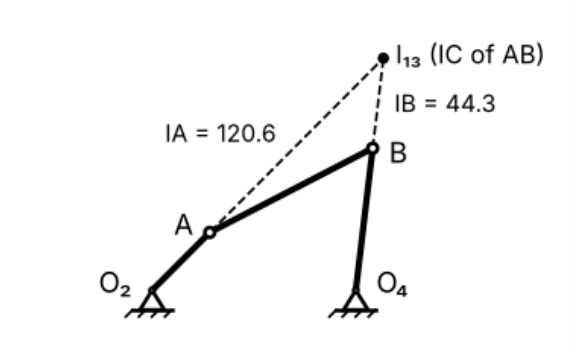
Coordinates (O_2 at the origin): A = (28.28, 28.28), B = (108.29, 69.51), O_4 = (100, 0)

Line O_2A (y = x) meets line O_4B at $\mathbf{I_{13} = (113.54, 113.54)}$, so IA = **120.57 mm** and IB = **44.34 mm**

$v_A = 15 \times 40 = 600$ mm/s → $\omega_{AB} = 600/120.57 =$ **4.976 rad/s**

$v_B = 4.976 \times 44.34 = 220.7$ mm/s → $\omega_4 = 220.7/70 =$ **3.15 rad/s**

Exam tip: Short cut: $\omega_4/\omega_2 = (O_2A/IA) \times (IB/O_4B)$. Always sketch the IC; it shows at a glance whether the answer is sensible.



Q8 **Answer: (A) perpendicular to the link, obtained by rotating the outward sliding velocity 90° clockwise**

Concept: Coriolis acceleration = $2\boldsymbol{\omega} \times \mathbf{v}_{rel}$. Its magnitude is $2\omega v$ and it is always perpendicular to the link.

Direction: rotate the relative (sliding) velocity vector through 90° in the same sense as ω.

ω is clockwise and v_{rel} is outward, so rotate the outward direction 90° clockwise.

Exam tip: The centripetal term ($r\omega^2$, toward the centre) acts along the link. The Coriolis term never does.

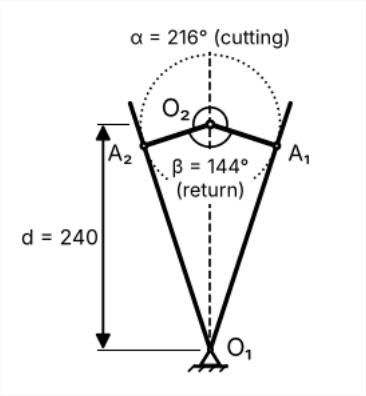
Q9 **Answer: 74.16 (accept 73.8 to 74.5)**

Concept: At the two extreme positions the slotted lever is tangent to the crank circle, so the crank is perpendicular to the lever. If β is the crank angle for the return stroke, cos(β/2) = r/d.

α/β = 1.5 and α + β = 360° → β = 360°/2.5 = **144°**, α = 216°

r = d cos(β/2) = 240 × cos 72° = 240 × 0.3090 = **74.16 mm**

Exam tip: The time ratio depends only on r/d. The lever length changes only the stroke length (2L·r/d).



Q10 **Answer: 532.96 (accept 530 to 536)**

Concept: Piston acceleration a = ω²r(cos θ + cos 2θ/n). At the dead centres this is exact: ω²r(1 + 1/n) at the IDC and ω²r(1 − 1/n) at the ODC.

ω = 2π × 900/60 = **94.25 rad/s**; ω² = 8882.6 s⁻²; n = 375/75 = 5

a_{ODC} = 8882.6 × 0.075 × (1 − 1/5) = **532.96 m/s²**

Exam tip: The IDC value is larger: 8882.6 × 0.075 × 1.2 = 799.4 m/s². That is why the maximum inertia force acts at the inner dead centre.

Q11 **Answer: 39 (accept 38.6 to 39.4)**

Concept: Velocity first, then acceleration: $a_B = a_A + \alpha_3 \times AB - \omega_3^2 AB$, and $a_B = \alpha_4 \times O_4B - \omega_4^2 O_4B$.

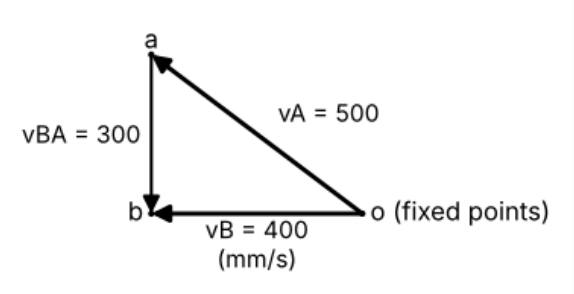
Velocity: $v_A = 10k \times (30, 40) = (-400, 300)$ mm/s. v_B is horizontal (O_4B is vertical), and AB is horizontal, so $300 + 100\omega_3 = 0 \rightarrow$ **$\omega_3 = -3$ rad/s** (cw). Then $v_B = (-400, 0) = -100\omega_4 \rightarrow$ **$\omega_4 = 4$ rad/s** (ccw).

$a_A = -\omega_2^2(30, 40) = (-3000, -4000)$ mm/s². $a_B = (-3000, -4000) + (0, 100\alpha_3) - 9(100, 0) = (-3900, -4000 + 100\alpha_3)$

From the rocker: $a_B = \alpha_4 k \times (0, 100) - 16(0, 100) = (-100\alpha_4, -1600)$

x-components: $-100\alpha_4 = -3900 \rightarrow$ **$\alpha_4 = 39$ rad/s²** (ccw); y-components give $\alpha_3 = 24$ rad/s².

Exam tip: Choose components that kill unknowns. Here the rocker is vertical, so the x-equation contains only α_4. Never drop the centripetal terms ($\omega^2 r$) in an acceleration polygon.



Q12 **Answer: (A) 48.2°**

Concept: Transmission angle μ = angle between the coupler and the rocker: $\cos\mu = (b^2 + c^2 - BD^2)/(2bc)$, where BD is the distance from the crank pin to the rocker pivot.

BD is smallest when the crank lies along the fixed link towards O_4: BD = 100 − 30 = **70 mm**

$\cos\mu_{min} = (90^2 + 80^2 - 70^2)/(2 \times 90 \times 80) = 9600/14400 = 0.6667 \rightarrow$ **$\mu_{min} = 48.2°$**

When the crank points away from O_4, BD = 130 mm and $\mu_{max} = 99.6°$.

Exam tip: Good designs keep μ between about 40° and 140°. The extremes always occur with the crank collinear with the fixed link.

Q13 **Answer: 5.77 (accept 5.71 to 5.83)**

Concept: The velocity of the crank pin A splits into a component along the lever (sliding velocity) and a component across it, which turns the lever: $\omega_{lever} = v_\perp / O_1A$.

A = (80 cos 30°, 200 + 80 sin 30°) = (69.28, 240) mm → $O_1A =$ **249.8 mm**; unit vector along the lever u = (0.2774, 0.9608)

$v_A = 25k \times (69.28, 40) = (-1000, 1732)$ mm/s (magnitude 2000 mm/s)

Across the lever: $v_\perp = v_A \cdot (-0.9608, 0.2774) = 960.8 + 480.4 =$ **1441.2 mm/s**

$\omega_{lever} = 1441.2/249.8 =$ **5.77 rad/s**; the sliding velocity is $v_A \cdot u = 1386.8$ mm/s

Exam tip: Both components are needed for the Coriolis term $2\omega_{lever} v_s$ (here 2 × 5.77 × 1.387 = 16.0 m/s²).

Q14 **Answer: (A) 110.8 mm**

Concept: The extreme slider positions occur when the crank and connecting rod are collinear: extended (distance l + r from O) and folded (l − r).

Outer: $x_1 = \sqrt{((l + r)^2 - e^2)} = \sqrt{(200^2 - 60^2)} =$ **190.79 mm**

Inner: $x_2 = \sqrt{((l - r)^2 - e^2)} = \sqrt{(100^2 - 60^2)} =$ **80.00 mm**

Stroke = 190.79 − 80.00 = **110.8 mm**

Exam tip: With an offset the stroke exceeds 2r and the two strokes take unequal crank angles, so the mechanism also gives a quick return.

Q15 **Answer: 0.848 (accept 0.839 to 0.856)**

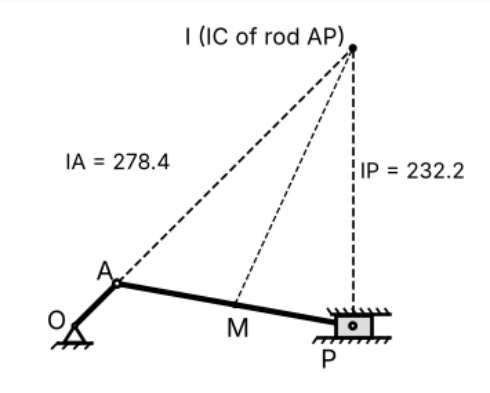
Concept: The IC of the connecting rod (I) lies where line OA extended meets the vertical through P (the perpendicular to the slider's velocity). Every point's speed is ω_{AP} × its distance from I.

A = (35.36, 35.36) mm; OP = 35.36 + $\sqrt{(200^2 - 35.36^2)}$ = **232.21 mm**; line OA is y = x, so **I = (232.21, 232.21)**

IA = 278.39 mm → $\omega_{AP} = v_A/IA = (20 \times 50)/278.39 =$ **3.592 rad/s**

M = (133.78, 17.68) → IM = **236.03 mm** → $v_M = 3.592 \times 236.03 = 847.8$ mm/s = **0.848 m/s**

Exam tip: The slider's speed is $v_P = \omega_{AP} \times IP = 3.592 \times 232.21 = 0.834$ m/s. A coupler point's speed is not the average of the speeds of its ends.



What next?

Day 14 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

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Next, Day 15/90: Engineering Mathematics, Calculus. Comment **GATE90** on our reel to get all links. Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.